

COURSE DESCRIPTION FORM

INSTITUTION FAST-NUCES

PROGRAM (S) TO BE EVALUATED BS (SE)

A. Course Description

Course Code	SE4001
Course Title	Software Re-engineering
Credit Hours	3
Prerequisites by Course(s) and Topics	Software Construction and Development (SE3001)
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	Assignments/ Quizzes – 10% Midterm – 30% Project – 10% Final Exam – 50%
Course Coordinator	Maqsood Khatoon (maqsood.khatoon@nu.edu.pk)
Class Teacher	Maqsood Khatoon (maqsood.khatoon@nu.edu.pk)
URL (if any)	-
Current Catalog Description	-
Textbook (or Laboratory Manual for Laboratory Courses)	Re-Engineering legacy software, David Lorge Parnas, Chris Birchall, Safari Books, Shelter Island, NY, 2016. S. Demeyer, S. Ducasse, and O. Nierstrasz. Object-Oriented Reengineering Patterns. Morgan Kaufmann, 2002.
Reference Material	Application Software Reengineering, Afshar alam, Tendai Patenga, Dorling Kindersley Pvt. Ltd, Pearson Education in South Asia, 2010. Software Maintenance and Evolution: a Roadmap, K.H.Bennett and V.T Rajlich, The Future of Software Engineering, ACM Press 2000.
Course Goals	The goals of the course are to teach the students the various concept, techniques, and methodologies used for software re-engineering at different phases of software development lifecycle.

A. Course Learning Outcomes (CLOs)					
No.	Course Learning Outcomes (CLO)	Domain	Taxonomy Level	PLO	Tools
01	Define the concepts and techniques of software reengineering. Demonstrate and utilize software re-engineering techniques to maintain and modify software systems.	Cognitive	2	2	A, M, P, F
02	Examine maintenance related problems associated with object oriented software systems. Investigate and design complex reengineering and reverse engineering problems	Cognitive	5	4	A, M, P, F
<i>Tool: A = Assignment,, P = Project, M = Mid-term, F=Final (End-term)</i>					

B. Graduating Attributes -GAs (Program Learning Outcomes – PLOs)			
1	Academic Education	Completion of an accredited program of study designed to prepare graduates as computing professionals	
2	Knowledge for Solving Computing Problems	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements	✓
3	Problem Analysis	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines	
4	Design/ Development of Solutions	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations	✓
5	Modern Tool Usage	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations	
6	Individual and Team Work	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings	
7	Communication	Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions	
8	Computing Professionalism and Society	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice	
9	Ethics	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice	
10	Life-long Learning	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional	

			Analysis, Integration and Customization, Recovering Lost Source Code, Fixing bugs and maintenance, Reverse Engineering Goals, Coping with Complexity, Recover lost information, Detect Side Effects, Facilitate Reuse	
	6.	3 hours	Restructuring in Re-Engineering, Code Refactoring, Architecture Enhancements, Removal of Obsolete Code, Sessional-I Exam, Reverse Engineering Tools, Code Refactoring, Architecture Enhancements, Removal of Obsolete Code	1
	7.	3 hours	Use Case Modeling in Reverse Engineering, Goals and Stories: Use Cases, Use Cases by Ivar Jacobson, Black Box Use-Cases, Domain Model, Characteristics of Domain Model, Features of Domain Model, Domain Classes, Entities, Attributes, Relationships, Aggregate, Composition	1
	8.	3 hours	Actions and Behaviors, Inheritance and Generalization, Association, Multiplicity, Role, Attribute, Class Visibility, Dependency and Generalization in Domain Model, Symbols of Relationships, Case study, Sequence Diagram, Sequence Diagram-Notations (Actors, Lifeline, Messages, Synchronous Messages, Asynchronous Messages, Create Message, Delete Message, Self Message, Reply Message, Found Message, Lost Message, Guards, Activation Bar, Alternative)	1
	9.	3 hours	System Sequence Diagram (SSD) vs Sequence Diagram, SSD- System Behavior, Re-Engineering Approaches, The Extent of Re-Engineering Performed, The Rate of Substitution of Operational System with New One, Re-Engineering Approaches (Big Bang Approach, Incremental Approach)	2
	10.	3 hours	Re-Engineering Approaches (Partial Approach, Iterative Approach, Evolutionary Approach), Source Code Reengineering Reference Model (SCORE/RM), The framework of Source Code Reengineering Reference Model (SCORE/RM), and its four kinds of elements: (Function, Documentation, Repository database, Metrication)	2
	11.	3 hours	Eight layers of the function element: (Encapsulation, Transformation, Normalization, Interpretation, Abstraction, Causation, Regeneration, Certification), Sessional II Exam	2
	12.	3 hours	Cyclomatic Complexity, Mathematical calculation of Cyclomatic Complexity, Control Flow Design, Design Patterns	2
	13.	3 hours	Design Patterns, Phase Reengineering Model, Analysis (Technical issues, Economic issue, and Planning (Understanding the scope of the work; Identifying the required resources; Identifying the tasks and milestones; Estimating the required effort; and Preparing a schedule), Phase Reengineering Model, Renovation Representation of the system, Representation of the external data),	2

			Target System Testing (Linear scripting, Modular, Data Driven)	
	14.	3 hours	High-level Reverse Engineering, Low-level Reverse Engineering, Redocumentation of artifacts, Design recovery, Pretty Printing, Static Analysis Tools, Dynamic Analysis Tools, Testing Frameworks, Key aspects of testing frameworks (Structure and organization, Automation, Reusability, Scalability), Redocumentation, Acceptance and System Transition	2
	15.	3 hours	Lexical Analysis, Deterministic Finite Automata, Token, Lexeme, Lexical Analyzer, Token classification, Token validation, Output generation	2
	16.	3 hours	Syntactic Analysis, Types of Parsing Algorithms used in Syntactic Analysis (LL parsing, LR parsing, LR(1) parsing, LALR parsing), Control Flow Analysis, Intra-procedural Analysis, Inter-procedural Analysis, Control Flow Graph (CFG), Data Flow Analysis, Basic Terminologies (Definition point, Reference point, Evaluation point), Data Flow Properties, Program Slicing	2
	17		Final Exam	1, 2
Laboratory Projects/Experiments Done in the Course				
Programming Assignments Done in the Course	Using UML for different diagrams, Reports & Presentations			
Class Time Spent on (in credit hours)	Theory	Problem Analysis	Solution Design	
	30%	35%	35%	
Oral and Written Communications	Students are required to design a project and submit 01 written project report of typically 08-10 pages and to make 01 oral presentations of typically 20 minute's duration. Include only material that is graded for flow of project, working of project, grammar, spelling, style, and so forth, as well as for technical content, completeness, and accuracy.			

- Usage of generative artificial intelligence (AI) such as ChatGPT or Microsoft Copilot for producing assessment submissions etc. is not allowed.

Instructor Name: Maqsood Khatoon

Instructor Signature: _____

Date: Jan 20, 2026